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Combustion Diagnostics and Emissions Measurements of a Novel Low NO_x Burner for Industrial Gas Turbine Operated With CO_2 Diluted Methane/Air Mixtures

While the employment of Exhaust Gas Recirculation (EGR) is a well-established technique in Internal Combustion Engines to limit NO_x emissions, its adoption in Gas Turbine engines has not yet found a practical application due to its expensive and complex installation that doesn't justify the emissions reduction when compared to already established DLN combustion technologies. EGR becomes an interesting option in GT engines considering the possibility of increasing the CO_2 content of the exhaust gases to improve the efficiency of Carbon Capture and Storage (CCS) units. However, the decrease in oxygen content of the combustion air is extremely challenging in terms of combustion stability and therefore of engine operability. In the present work, a low NO_x burner was studied at ambient pressure in a reactive single burner test rig. The burner was fed with methane and characterized in terms of emissions and stability limits at different operating conditions. In addition, the flame position and shape were studied through OH chemiluminescence imaging together with the flow field thanks to PIV measurements. The effects of CO_2 addition on the flame were then investigated at different EGR increasing levels, highlighting the impact of the oxygen content on the combustion reaction intensity. Variations in emissions and burner stability limits in terms of maximum sustainable CO_2 content were also studied, to detail the burner operating window. Data have been thoroughly analyzed to gather information on the burner behavior to support the design of new technical solutions capable of ensuring both proper flame stability and low CO and NO_x emissions.* [DOI: 10.1115/1.4064266]

Introduction

The continuous growth in energy demand will push the energy production toward more sustainable, flexible and highly efficient solutions, driving the greenhouse gases emission abatement and limiting the negative effect on climate change.

Natural gas employment in the energy sector is expected to increase thanks to its essential system flexibility [1]. In advanced economies, it represents the largest source of electricity and its share is going to remain stable over the next decade despite the emerging renewable sources. Therefore, increasing the use of Carbon Capture and Utilization (CCU) and Carbon Capture and Storage (CCS) technologies is of fundamental importance to reduce greenhouse gases emissions. In order to reduce the cost of CCS implementation and to create an efficient system, the CO_2 content at the inlet of the carbon capture unit must be maximized through the adoption of high percentages of Exhaust Gas Recirculation (EGR). The EGR employment as a mean to control the flame temperature and limit NO_x emissions is a common practice in internal combustion engines. However, due to the high costs and complexities related to its installation, it hasn't been considered for real applications to gas turbine engines. Indeed, the achievable reduction in NO_x emissions doesn't justify the power plant complexity, especially if compared to the performance of modern Dry Low NO_x (DLN) combustion technologies which are usually able to provide close to single digit emissions [2]. Thus, the most interesting EGR application to gas turbines is related to the achievable increase of CO_2 content in the exhaust gases which would benefit the CO_2 sequestration both in terms of costs and efficiency. On the other hand, such operating

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conditions push conventional combustion systems to the limit because of oxygen depletion.

Over the years, several studies have investigated the possibility of combining EGR with lean premixed flames in order to ensure the proper flame stability together with low CO/UHC (unburnt hydrocarbons) and NO_x emissions. In terms of NO_x emissions, EGR showed promising results. At constant flame temperature and high pressures, the EGR NO_x reduction potential increases as the amount of recirculating gas rises [3]. In addition, compared to standard air cases, the NO_x emissions reduction is even bigger at increasing flame temperatures [4]. The reason behind this behavior is related to the driving NO_x formation mechanism: EGR NO_x potential reduction is associated with the Zeldovich NO_x formation pathway, thus the reduction is effective only if this is the dominant formation mechanism, so in high pressures and high temperatures conditions [5]. This aspect was additionally investigated by several authors that not only confirmed this behavior but also found an inversion in the NO_x trend for extremely high EGR levels [6]. This NO_x trend inversion phenomenon was also investigated through numerical simulations by Burnes et al. [7], who found a threshold in terms of oxygen content of the incoming air above which the NO_x emissions starts rising again.

The real limiting factor associated with the EGR employment is related to CO emissions [8]. The oxygen depletion caused by an increased EGR level drifts the thermodynamic equilibrium toward the CO production. Nevertheless, the CO emissions linked to the equilibrium conditions are only a theoretical value, which is expected to deviate noticeably as the lack of oxygen in the fresh mixture increases [3]. Additionally, the increase in CO emissions is supposed to be particularly pronounced at low loads engine operating conditions, where the lack of oxygen effect overlaps with a decrease in pressure. Different strategies have been explored in order to extend the combustor operability window: the majority of researchers studied the employment of a small pilot flame to increase the flame stability at high EGR content [8,9], while others explored the possibility of employing staged combustion [3,4]. The EGR impact on the combustion process has been mainly investigated employing numerical methods, most of which are usually limited to low fidelity approaches such as simple one-dimensional studies [10–12] or Reynolds-averaged Navier Stokes (RANS) simulations [13–16]. Although Large Eddy Simulations (LES) would be preferable in such conditions to provide a detailed understanding of the interaction between the turbulent flow field and the flame characteristics, the available applications are limited to simple combustor configurations due to the modeling high computational costs [17–20]. Dedicated experimental investigations are required in order to support the prediction of numerical models, and combustion optical diagnostics have been successfully employed in this context. A limited number of available studies investigated at atmospheric pressure the effects of CO_2 addition on the flame behavior through chemiluminescence measurements, yet considering simple burner geometries [21,22].

The main objective of this work is therefore to contribute to the development of new burner concepts, able to efficiently operate in alternative operating conditions such as exhaust gas recirculation. Specifically, optical diagnostics measurements have been performed on an industrial burner at ambient pressure conditions to investigate the complex behavior of real hardware components. To this end, a first test campaign is dedicated to the burner characterization while the second part is focused on the assessment of the actual capabilities of the currently employed burner, designed to operate in standard gas turbine cycles.

After characterizing the flow field in both nonreactive and reactive conditions, the burner flame structure and position were investigated through OH* chemiluminescence measurements. In addition, NO_x and CO emissions were recorded at the same time to characterize the burner performance in terms of pollutant emissions. For a complete characterization, the burner was also studied from a stability point of view through lean blow out (LBO) measurements.

The measurements will be used to study the involved combustion phenomena and the complications related to the combustor operability in vitiated air conditions, identifying the limitations of the current technologies and testing the burner ability to adapt to extreme operating conditions. In addition, the collected information represent a good database to be employed for CFD codes validation.

Test Rig and Experimental Techniques

Burner Description. The investigated burner is a novel lean premixed burner developed by Baker Hughes for industrial gas turbine applications. A thorough description of the burner geometry and its design can be found in the works of Cerutti et al. [23–25]. The burner cross section is illustrated in Fig. 1.

The burner mainly consists of two different parts:

- Part 1 of Fig. 1 is the fuel injector, equipped with two independent fuel injection circuits
- Part 2 of Fig. 1 is the premixer, composed by two independent air flow passages, each one providing a different amount of swirl to the air-fuel mixtures

Each one of the fuel injection rows is independently fed and regulated by a different fuel gas line. According to Fig. 1, a first air stream AA (Axial Air) enters the burner through an axial swirler and mixes with the first row of axial fuel injectors AF (Axial Fuel) present within the external passage, just upstream the axial swirler, creating a premixed flame. The second air stream RA (Radial Air) flows through the internal vanes of the radial swirler before mixing with the radial fuel stream RF (Radial Fuel) to feed the internal premixed flame. The following throat is properly sized to ensure an appropriate mixture acceleration before the downstream sudden section expansion, which corresponds to the flame front developing region. Regardless of the amount of residual flow swirl, the flame stabilization mechanism is mainly driven by the inner throat size and the law of cross-sectional area increase of the downstream dump region. As pointed out from previous investigations [23–25], the flame stabilization process is also strongly affected by the amount of overall thermal power released in each region through the dedicated radial and axial fuel injections.

Test Rig Configuration. The burner was investigated in the reactive test cell of the THT Lab of the University of Florence, whose schematic layout is outlined in Fig. 2. Two screw compressors deliver a maximum of 1 kg/s of compressed air to the test cell at a maximum pressure of 10 barA. A modulating valve is also integrated in the system to ensure a constant delivery pressure. After the compression, the air flows through various filters and an air dryer (dew point −20 °C) in order to eliminate any impurity and humidity before entering the test cell. Right after, a 600 kW electric heater heats up the air flow at a maximum temperature of 400 °C. To keep a constant delivery temperature, a proportional integral derivative system automatically controls the electric heater, modulating its power input.

The air mass flowrate is controlled through a globe valve upstream of the electric heater, while the rig downstream pressure is set acting on a high temperature control disk valve. A water jacket exhaust duct is placed right after the rig to avoid any interference the

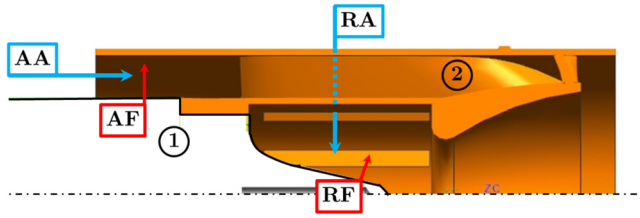

Fig. 1 Investigated burner architecture © 2023 Baker Hughes Company - All rights reserved

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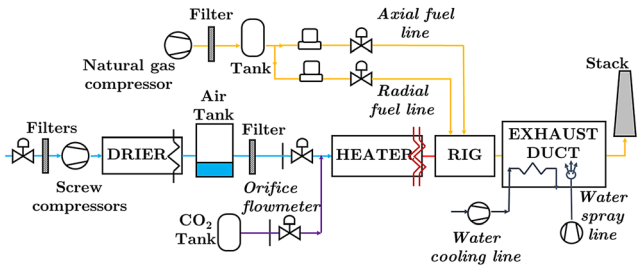


Fig. 2 Combustion test cell sketch

back-pressure valve might have on the combustion process. The water jacket is a 1 m long straight duct with circular cross section equipped with water sprayers, used to quench the hot gases and control the flow exit temperature.

The burner has been tested both in standard air conditions and CO_2 enriched mixtures, to simulate EGR like conditions. Recreating real EGR conditions in lab scacle is quite challenging because of the associted costs and the volumes involved; the same complications are when employing N_2 enriched mixtures. Therefore, CO_2 enriched mixtures are considered a good tradeoff for the design and primary investigation phase. The CO_2 employed for the tests in vitiated air conditions is stored in a pressurized tank with a storage capacity of about 6000 l. From the storage tank, the CO_2 line directly enters the main air flow line upstream the electric heater, in order to deliver to the test section an homogeneous mixture, both in terms of temperature and composition. The amount of injected CO_2 is controlled through a dedicated valve and measured by a calibrated orifice flowmeter.

As for the fuel line, the test cell is equipped with three natural gas lines that deliver a maximum flowrate of about 90 Nm^3/h. The natural gas directly coming from the local gas network is pressurized from 20 mbar to about 16 bar through a dedicated screw compressor, equipped with special filters to clean the gas from possible impurities. A 200 liters tank is also present along the delivery line to smooth out any pressure fluctuation. Each fuel line is provided with a globe valve to control the flowrate and a shut off valve for safety reasons, together with a Pt100 resistance temperature detector probe, pressure gauges, and a specific mass flowmeter.

The test article is designed to allow optical measurements on the flame region and it is characterized by a modular layout to adjust the system to different burner concepts. A cross section of the test article is shown in Fig. 3(*a*). It consists of two main parts: the casing, which constitutes the enclosure that confines the air and the reactive process ensuring a safe test operation, and the combustor, which is the flame enclosure system. The external casing consists of an upstream vessel, through which the burner enters the test article, and a downstream vessel, equipped with two perpendicular optical windows for the flame visualization. The combustor is made of a 2.5 mm thick cylindrical quartz liner, held in place thanks to a blocking system made of four tie rods and four springs that connect the dome to the following duct, used to convey the exhaust gases toward the test rig exhaust system. The combustor length is approximately 2.5 diameters, to ensure a complete flame development for a proper flame visualization. In addition, special effort was put in the optical accesses design to ensure a good visibility on the dome and burner exit section, as well as the combustor terminal part, to investigate any interaction it may arise between the flame and the dome surface.

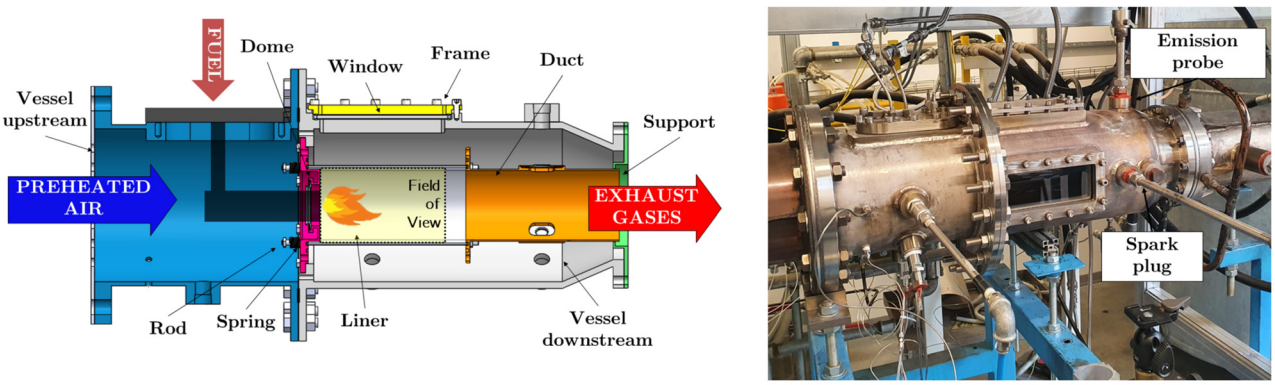
The flame tube is cooled by forced convection of a fraction of the incoming air that flows in the annulus between the liner and the confining vessel, not used in the combustion process. The same air is also employed for the windows cooling, while the dome plate is cooled through a series of inclined effusion holes. Dedicated flow checks were performed to estimate the effective areas of both the burner and the cooling system, to allow the actual combustion air feeding the burner to be evaluated.

In order to perform tests in reactive conditions, the rig is provided with a spark plug to ignite the mixture, located along the flame tube downstream the optical liner (Fig. 3(*b*)).

Concerning the rig instrumentation, the test article is equipped with various thermocouples and static pressure ports to monitor the flow conditions. An emission probe is also present, employed to analyze the exhaust composition through a HORIBA PG250. The probe is made of several radially spaced holes and it is plunged into the flame tube to extract the exhaust gases. After being extracted, the gases reach a gas analyzer through a thermally insulated 6 mm teflon pipeline that keeps the gas mixture at a constant temperature of about 150 °C. Before reaching the analyzer, a HORIBA PS200 refrigerator is used to remove any trace of water vapor from the gas sample. The gas analyzer is properly calibrated right before each test employing a rack of calibrated gas mixture tanks.

An in-house LabVIEW software tool is used for real-time monitoring of all the main rig parameters. The acquired data at each test point are obtained by averaging the measured values for 30 s, once reached the desired stationary conditions. A picture of the test rig and the relative instrumentation is showed in Fig. 3(*b*).

Particle Image Velocimetry. The PIV investigations were carried out employing the Dantec-Dynamics PIV system, where the © Dantec-FlowManager software was used to control all the instrumentation settings, as well as synchronization, acquisition and postprocessing. The laser adopted in this test campaign is the Litron LDY303, a Nd:YAG pulsed laser at a wavelength of 527 nm. The laser beam is shaped into a thin laser sheet thanks to an optical trail, made of a series of lenses. A final mirror mounted on an adjustment system is used to project the laser sheet into the test rig from the



(*a*) (*b*)

Fig. 3 Test article cross section (*a*) and combustion test rig photo (*b*) at the THT Lab of the University of Florence

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upper window (Fig. 3(*a*)) and illuminate the central section of the liner, at the burner symmetry plane.

The images were recorded through a Phantom MIRO M340 camera, coupled with a Micro-NIKKOR 60 mm *f*/2.8 lens and a bandpass filter to cut out all the incoming light except for the laser light scattered by the seeding particles. A polarizing filter is also placed in front of the bandpass filter of the camera lens to better manage the reflections caused by the cylindrical liner shape above all. TiO_2 particles (submicron mean size) were used to seed the air flow directly upstream the burner, using the Lavision Particle Blaster 110. Both cold and reactive tests were conducted acquiring images at a frequency of 400 Hz and the time delay between the two laser pulses was varied between 7 and 9 μs. For each tested condition, about 500 pair of images were acquired to assure statistical convergence of the results. Background noise was eliminated through a dedicated algorithm and the data were then postprocessed using an adaptive grid iterative approach. Vectors were validated applying peak validation on the image cross-correlation and the Universal Outlier Detection algorithm [26].

OH* Chemiluminescence. Chemiluminescence measurements of the OH* radical were employed to detect the reaction zone and its position in various operating conditions, as its emission can be considered a heat release marker [27–29]. In order to carry out this type of measurements, a camera coupled with an image intensifier has been used. The same camera used for the PIV measurements, the Phantom MIRO M340, was coupled with the Hamamatsu C16031-311-Ax image intensifier through a relay lens. In addition, a Nikon UV 105 mm $f/4.5$ lens together with a bandpass filter (CWL = 310 +/− 2 nm; FWHM = 10 +/− 2 nm) were mounted on the image intensifier to be able to capture the OH* transition, which has its peak emission intensity in the UV spectrum. OH* chemiluminescence images were acquired at 500 Hz recording about 900 images. The instantaneous snapshots were averaged to remove the influence of any unsteady phenomena and analyze the flame steady-state structure. In addition, the averaging process smooths out the chemiluminescence intensity peaks associated with unsteady structures and reduce the measurement noise. Throughout this test campaign, the intensifier gain was kept constant while the gate was varied between 0.4 and 0.7 ms depending on the flame luminosity (, i.e., tested conditions). For the cases acquired with different intensifier gate values, each averaged test condition was normalized over its maximum intensity value. Thanks to this operation, the aforementioned cases were compared in terms of flame shape and position and extension of the reaction zone.

Tested Operating Conditions. The burner was tested in different operating conditions in order to study the flame changes in shape and position through combustion diagnostics techniques. The operating conditions are determined through the following main control parameters:

- T_i is the combustor inlet temperature, set to 300 °C to simulate typical gas turbine compressor discharge temperatures
- P_i is the combustor air inlet pressure. As early mentioned, the test campaign was conducted at ambient pressure
- $\Delta P/P$ is the pressure drop across the combustion chamber, calculated as the differential pressure across the combustor divided by the combustor inlet pressure. All tests were performed at a ΔP of about 4.5% which was found to provide stable low emission operating window with natural gas [23,25]
- $T_{flame}*$ is the flame temperature, calculated as the adiabatic flame temperature assuming a perfect mixing between the burner combustion air and the total amount of fuel injected. The flame temperature values are scaled with the typical ISO base load flame temperature for proprietary reasons ($T_{flame}/T_{flame,ISO}$)
- RF, the fuel split between the radial and the axial fuel lines, which is expressed as percentage of the total fuel amount per each fuel injection row

Table 1 Burner operating conditions for the OH* chemiluminescence measurements

Case	$T_{\text{flame}}*$	Fuel split (%)	O_2 (%mol)
1	0.75	100% RF	21
2	0.86	100% RF	21
3	1	100% RF	21
4	1	60% RF–40% AF	21
5	1	50% RF–50% AF	21
6	1	40% RF–60% AF	21
7	1	100% RF	19.5
8	1	100% RF	19
9	1	100% RF	18.5
10	1	50% RF–50% AF	20
11	1	50% RF–50% AF	19.5
12	1	50% RF–50% AF	19
13	1	50% RF–50% AF	18.5
14	1	40% RF–60% AF	20
15	1	40% RF–60% AF	19.5
16	1	40% RF–60% AF	19
17	1	40% RF–60% AF	18.5

- The air vitiation in close-to EGR conditions is expressed as oxygen molar concentration (O_2%), reached after the addition of a certain amount of CO_2 into the main air flow entering the test rig ($O_2 = 21\%$ for standard air).

Table 1 summarizes the tested operating points. The axial fuel line was not fed further to not exceed the 40% RF–60% AF fuel split configuration, as it was selected as the most favorable in terms of NO_x emissions during the burner development process [23–25]. In addition, the burner stability limits were investigated both in standard and vitiated air conditions.

Burner Characterisation

Flow Field Results. The mean flow field has been measured on the burner symmetry plane with PIV technique in both reactive and nonreactive conditions. Figure 4 shows the time averaged velocity magnitude of the nonreactive (up) and reactive (down) flow fields, normalized with the maximum velocity value of the reactive case. For the reactive test, only the radial fuel line is fed and T_{flame} is equal to 0.75, corresponding to case 1 of Table 1. The white solid lines highlight the typical combustion chamber flow field for a swirl stabilized flame, and flow vectors are also reported. In both conditions, the swirling motion generated by the swirlers forces the flow to spread radially toward the chamber liner wall, creating a low velocity outer recirculation zone (ORZ) in the corner between the dome and the liner. The sudden expansion of the flow causes an

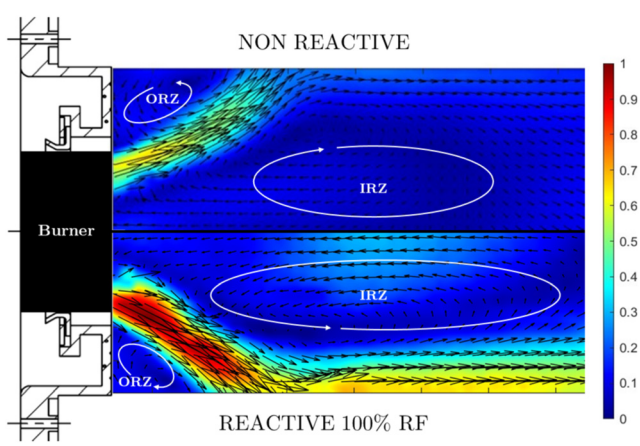

Fig. 4 Normalized mean velocity contour with a schematic of the typical flow field and flow vectors in nonreactive conditions (up) and reactive conditions (down)

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increase in the vortex size and consequently a pressure rise in the vortex core, generating a positive pressure gradient in the central region of the combustion chamber. The flow field is therefore subject to a vortex breakdown, characterized by a negative axial velocity in the central region of the combustion chamber [30,31]. As a result, the flow develops an inner recirculation zone (IRZ) at the central part of the chamber, highlighted by the solid white line of the picture. This region is fundamental for the flame stabilization, as it continuously supplies heat and activated radicals from the hot combustion products toward the fresh mixture. In the area bounded by the outer and inner recirculation zone, the flow is accelerated and forms a shear layer between the high and low velocity regions generating high turbulence to enhance the mixing. The IRZ root and its initial interaction with the jets is not visible as it is anchored in the burner final dump region, close to the burner exit section. Similarly, it was not possible to collect information on the IRZ final part as it is located outside of the camera FoV, especially in reactive condition where the extension of the inner recirculation zone gets slightly larger.

Clearly, there is no big difference between nonreactive and reactive conditions in terms of flow field features. The combustion process, thus the increase in temperature, leads to higher velocities. Furthermore, the thermal expansion and the lower densities of the reactive conditions cause an increase in the extension of the IRZ, especially in the radial direction. Also, in the reactive case, the flow entrapped in the inner recirculation zone moves back toward the burner with higher velocity compared to the nonreactive case. This aspect together with the resulting larger extension of the IRZ forces the incoming high velocity jet outwards with a steeper angle. Consequently, the ORZ shortens and it is pushed closer to the burner wall.

Nonreactive and reactive flow field measurements were also performed at higher T_{flame} and with different fuel splits, but the results are very similar to those presented above and have been omitted for brevity.

OH* Chemiluminescence Results. The burner flame structure and position were investigated through OH* chemiluminescence measurements at different operating conditions in terms of flame temperature and burner fuel flow split. Figure 5 presents all the time averaged OH* maps recorded with standard air (, i.e., without CO_2 addition), corresponding to cases 1–6 of Table 1. Each image is normalized with its own maximum value.

The effect of flame temperature can be studied considering images a-b-c, corresponding to cases 1–2–3, where the fuel flow split was kept constant, feeding only the radial fuel line. The main reaction zone can be identified through the high OH* chemiluminescence values. The flame temperature is raised by increasing the amount of fuel that flows through the radial fuel line, leading to a richer mixture. Although the equivalence ratio remained well below the stoichiometric value for all the considered cases, the increase of the jet penetration forced the OH* peak relocation downstream. In particular, as the amount of fuel is increased it is supposed that a larger portion of that interacts with the high velocity swirled air, determining a heavy lifting effect and modifying the flame stabilization region (Fig. 4). As the jet penetration increases, the flame stabilizes further downstream the burner exit section assuming the lifted shape visible in Fig. 5(*c*). As a result, the peak of chemiluminescence intensity moves accordingly and it gets closer to the liner walls. From case 1 to case 3, the shape of the reaction zone first gets slightly larger and bigger and then changes shape, becoming more squeezed and moving away from the burner tip.

To gain a better understanding of the flame anchoring position, OH* chemiluminescence map of case 1 is overlapped with PIV velocity vectors measured at the same operating conditions, as showed in Fig. 6. Since chemiluminescence is a line-of-sight technique, the inverse Abel transform [32] has been applied to the time averaged OH* map, recreating the intensity distribution on the burner symmetry plane. The flame clearly anchors in the inner shear layer region where high turbulence is promoted through the interaction between the IRZ and the jet flow exiting the burner.

The effect of the burner fuel split between the axial and the radial lines on the flame structure was also investigated. Figures 5(*c*)–5(*f*) were obtained keeping the flame temperature constant at 1 and

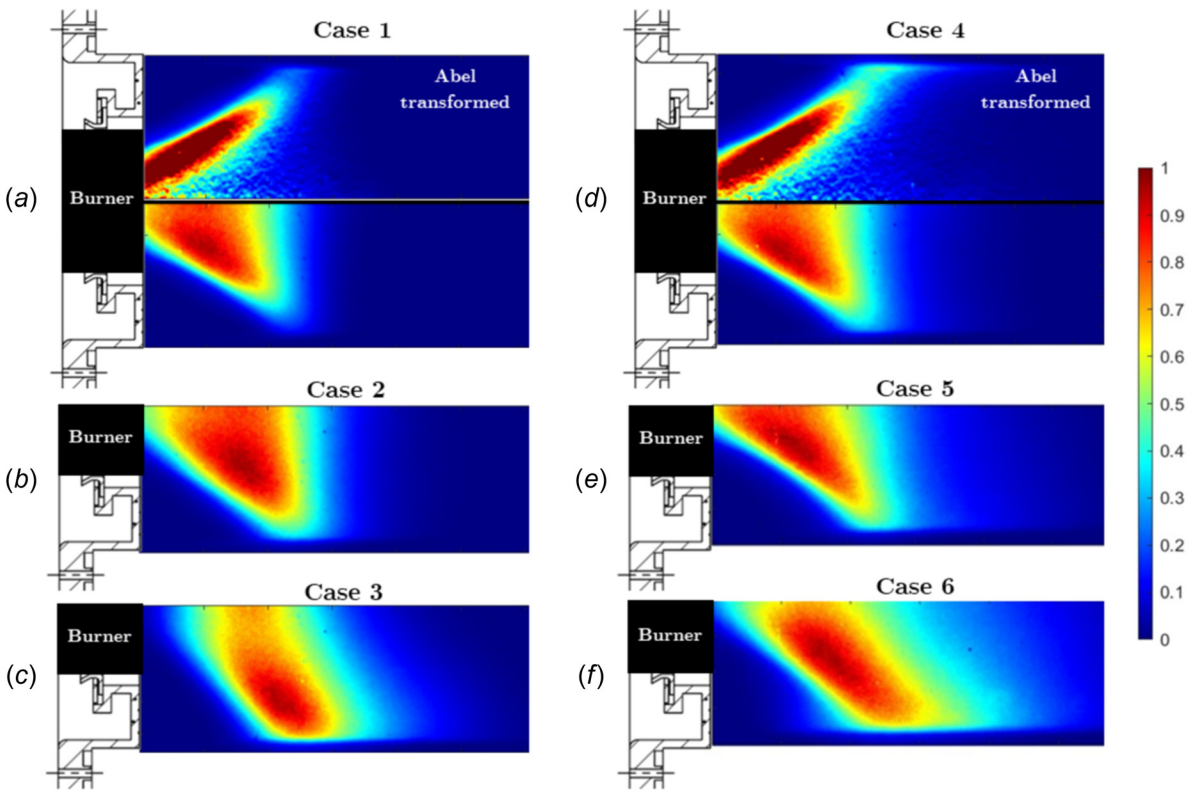

Fig. 5 OH* chemiluminescence contours

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progressively decreasing the radial line content from 100% to 40% and increasing the axial fuel line consequently. As the amount of fuel flowing through the radial line diminishes, the flame anchors back in the central region closer to the burner exit. This behavior endures until the last tested fuel flow split, where the amount of fuel through the axial fuel line is greater than the radial one. This behavior is driven by the fuel jet velocity of the radial line. When the total fuel amount flows through the radial line, the fuel velocity is higher than the fresh air thus the flame stabilization happens downstream the burner exit section, assuming a lifted shape. In Figs. 5(*d*) and 5(*e*), the fuel stream through the radial line decreases and its exiting velocity is low enough for the fuel to get entrapped in the recirculation zone created in the dump region of the burner (Fig. 1). Indeed, the channel throat after the radial line location provide the mixture with the proper acceleration through the downstream sudden section expansion, where the flame then anchors. However, in the last case the radial line equivalence ratio gets leaner and the flame behavior starts shifting back toward a lifted configuration. Thus, in Fig. 5(*f*) the reaction zone lengthens, relocating the flame a little downstream. Also, the chemiluminescence intensity peak moves away from the liner wall as the radial line fuel starts decreasing and start shifting again toward the liner only in the last case. Actually, the flame shape of Figs. 5(*c*) and 5(*f*) are quite similar: the reaction zone gets shorter as it spreads more radially, interacting more with the combustion chamber walls.

Emissions. Figure 7 show the CO and NO_x emissions as a function of the normalized flame temperature at constant fuel split (cases 1–3). As expected, CO emissions decrease at higher flame temperature. Recalling the chemiluminescence maps of Fig. 5, it is clearly visible that at leaner conditions the flame gets closer to the burner exit section, creating an initial richer zone. In this area, the poor fuel-air mixing is responsible for the evident increase in CO emissions. At higher flame temperature, even though the flame interacts with the liner wall that cools down the mixture blocking the reaction progress, the flame anchors in a different position and the better mixing contributes to the CO emissions abatement. The trend of NO_x emissions remains almost flat, assuming almost a constant value around 11 ppm for the lowest and highest flame temperature, while a slight increase in NO_x emissions is visible for the medium case, ascribable to the slight reactivity increase found in the OH* analysis.

The emissions trends were also analyzed at various fuel split conditions at constant adiabatic flame temperature (cases 3–6). Figure 8 illustrates the CO and NO_x emissions as a function of the radial fuel line split. As it could be expected, the maximum CO

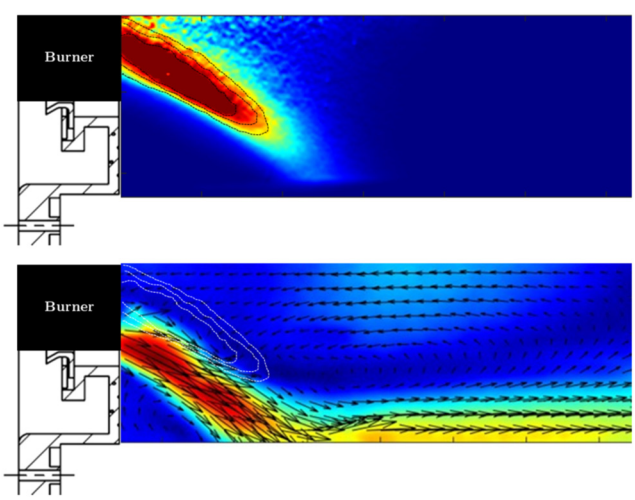

Fig. 6 Abel transformed OH* chemiluminescence intensity isolines of Case 1 superposed by the velocity contour with flow vectors of the PIV measurement

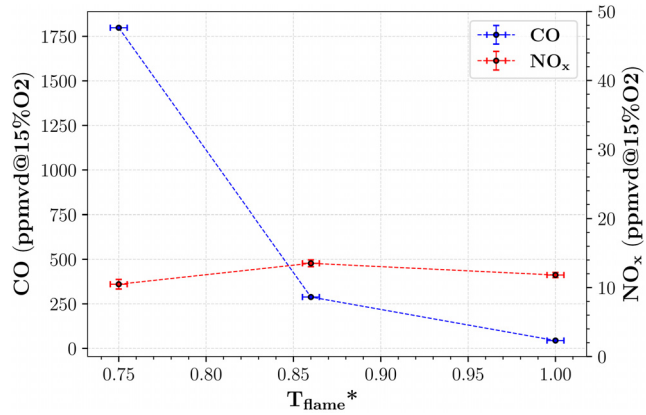

Fig. 7 CO and NO_x emissions as a function of the adiabatic flame temperature at 100% RF

emissions is found at 100% RF. As it was explained for the trend obtained at different adiabatic flame temperatures, looking at the chemiluminescence of Figs. 5(*c*)–5(*f*) it is possible to see the change in shape that the flame experiences from 100% to 40% RF. Moving from case 3 to case 4, there is an initial marked drop in CO emissions due to an enhanced mixing promoted by the axial stream. A further increase in the axial fuel line content still benefits CO emissions but at a lesser extent and the trend assumes an asymptotic behavior. Similarly, NO_x emissions decrease as the AF contribution grows. The favorable effect of the better mixing linked to the activation of the axial fuel line can be appreciated already from small amount of AF content, leading to a reduction in NO_x emissions by a factor of 4 from 100% RF to 60% RF. The trend then stabilizes and reaches a minimum at 40% RF. Therefore, a fuel split biased toward the axial fuel line reduces the radial line equivalence ratio, which is responsible for the majority of the NO_x emissions formation. However, as the radial line is mainly responsible for the flame anchoring and stabilization, a balance must be found to ensure a stable combustion at the same time.

As mentioned before, the radial fuel line contribution was not decreased further since from previous investigations an increase in NO_x emissions is expected at an axial fuel content greater than 60% as well as a rise in thermoacoustic instabilities, which exclude this region from the burner desirable operating conditions [23–25].

Lean Blow Out Margin. One of the most important aspects in a burner design is its capacity to provide stable combustion. The burner stability limits define its capacity to withstand variations in terms of flame temperature without the onset of combustion instabilities or loss of flame events. These limits become critical

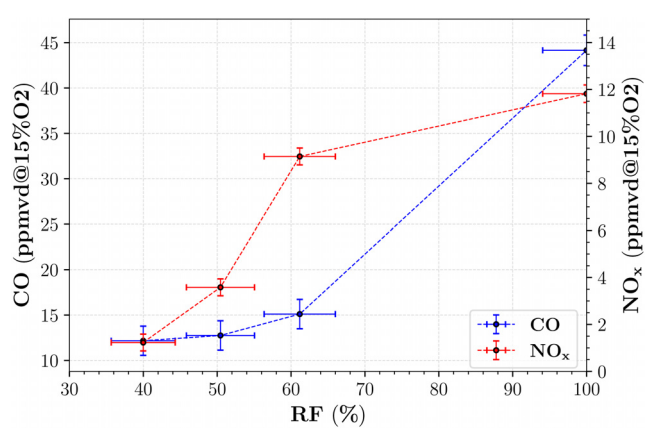

Fig. 8 CO and NO_x emissions as a function the amount of fuel flowing through the radial fuel line (RF) at constant flame temperature ($T_{flame}^* = 1$)

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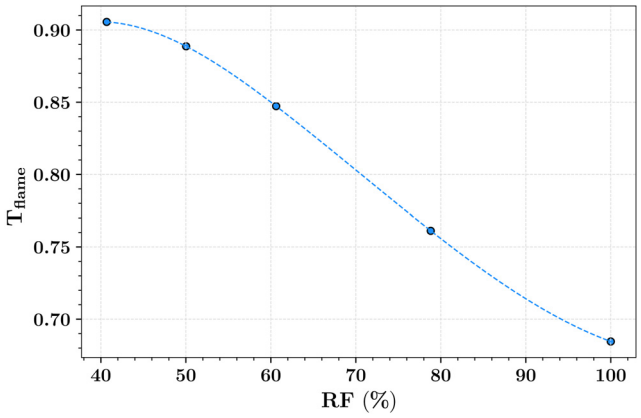

Fig. 9 Lean blow out measurements expressed as corresponding adiabatic flame temperature as a function the amount of fuel flowing through the radial fuel line (RF)

especially at high axial fuel content conditions. The blow out margin definition is fundamental particularly during the gas turbine load regulation, as often gas turbines are required to follow the frequent variations of the grid power demand in an energy context where the energy production share of renewable sources, characterized by an irregular availability, is in continuous growth. In addition, wider stability limits allow the combustor to operate in leaner conditions, reducing thus the adiabatic flame temperature while keeping low NO_x emission capability.

Lean blow tests were carried out to determine the burner stability limits in different radial-axial split configuration, namely, cases 3–6. For each case, starting from stable conditions, the air mass flowrate was gradually increased while maintaining the pressure drop across the combustion chamber within an acceptable range, until the flame extinguishes. The blow out event was detected observing the sudden temperature drop from the thermocouples readings, installed in the duct close to the liner exit section. For each test point, the blow out test was performed three times showing a good repeatability.

Figure 9 shows the burner stability limits in terms of adiabatic flame temperature (T_{flame}) as a function of the radial fuel split (RF %). Looking at the two limiting conditions, for the 100% RF case the flame remained stable until the adiabatic flame temperature reached a value below 0.65. Differently, only a T_{flame} of 0.85 was reached for the 40% RF case. Indeed, as the axial line content increases, the blow out margin decreases and the burner moves to a more unstable condition. As derived from the chemiluminescence measurements, the radial line is responsible for the flame stabilization so when its contribution diminishes the flame is less stable and the blow out margin decreases accordingly. Indeed, going from case 3 to case 6, the raw OH* luminosity experience a considerable reduction indicating a lower flame reactivity that negatively affects the flame stability.

As seen in the emission section, case 6 which corresponds to the minimum investigated RF content is the most promising in terms of NO_x emissions but on the contrary it is the worst configuration in terms of stability limits. The axial line globally provides a better mixing achieving a more uniform fuel distribution, which is an important aspect in terms of emissions but at the same time it is detrimental for the flame stability as richer regions are responsible for the flame stabilization. The drop in NO_x emissions at around the 50% RF–50% AF fuel split and the corresponding almost linear behavior of the lean blow margin offers a promising opportunity in finding a tradeoff between these two characteristics.

Effect of CO_2 Dilution

The present study aims at testing the burner adaptability at different operating conditions and understands the potential modification needed to operate in an oxygen depleted environment. As explained before, CO_2 was added into the main airflow to simulate EGR conditions. The CO_2 content was gradually increased to decrease the oxygen available for the combustion process. Four different operating conditions in terms of available oxygen concentration in the incoming airflow were tested, as reported in Table 1. In order to derive a truthful comparison on the burner behavior, the adiabatic flame temperature was kept constant at all conditions adjusting the fuel flow. The air mass flowrate was also varied depending on the added CO_2 in order to maintain the right burner pressure drop.

Stability Limits. One of the principal issues related to the EGR feasibility is the flame stability. Indeed, the addition of CO_2 into the incoming airflow reduces the amount of available oxygen for the combustion process, weakening the flame stability.

Therefore, the flame blow out margin was studied in order to derive the maximum allowable CO_2 concentration to ensure stable combustion in different fuel flow split configurations. Similarly to the procedure adopted for the burner characterization, for each tested condition, the fuel flow was kept constant, and the CO_2 content was increased until the flame extinguished. Figure 10 shows the burner blow out margins in terms of oxidant oxygen content as a function of the fuel flow split, expressed in terms of radial line content. As the radial fuel flow diminishes, the minimum oxidant oxygen content increases to ensure flame stability. Indeed, as shown in the burner characterization both from chemiluminescence measurements and stability limits, the radial fuel helps the flame anchoring and stabilization process. Thus, at 100% RF, the minimum oxygen molar concentration in the oxidant mixture reaches the lowest experienced value at about 18%. The oxygen molar concentration necessary for a stable combustion increases as the axial fuel stream grows, reaching a value of about 18.45% for the 40% RF - 60% AF fuel split case. Generally, the needed oxygen content rises with a strong nonlinear behavior as the radial fuel content decreases, with a markedly increase in the needed oxygen content around the value of 50% RF.

This information was used to define the most interesting configurations to be further analyzed in terms of flame shape through OH* chemiluminescence and pollutant emissions measurements. As pointed out previously, the most unstable condition is the one at greatest axial line content and it drives the maximum CO_2 oxidant content tested. Therefore, to ensure stable combustion, a 18.5% oxygen molar concentration was chosen as lowest allowable oxygen oxidant content to be investigated during the burner characterization in vitiated air conditions.

OH* Chemiluminescence Results. Figure 11 presents OH* chemiluminescence intensity averaged over the whole measurement area for different fuel splits at decreasing oxygen content. The intensifier gate was kept constant with the purpose of illustrating the

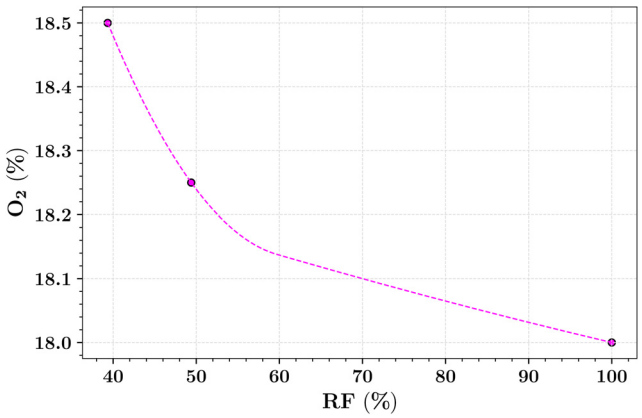

Fig. 10 Lean blow out measurements expressed in terms of oxidant oxygen content as a function of the amount of fuel flowing through the radial fuel line (RF)

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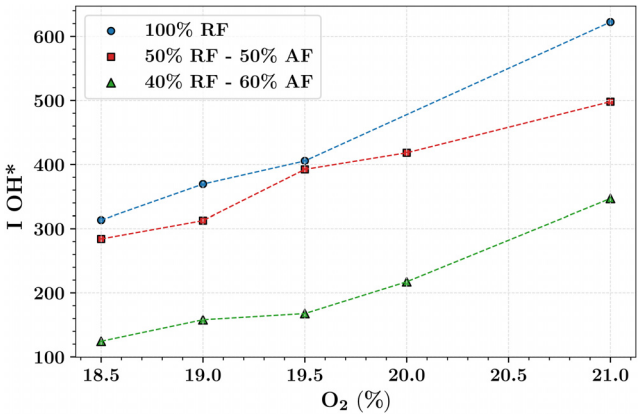

Fig. 11 Average OH* chemiluminescence intensity for different fuel splits at decreasing oxygen content and constant flame temperature ($T_{flame}^* = 1$)

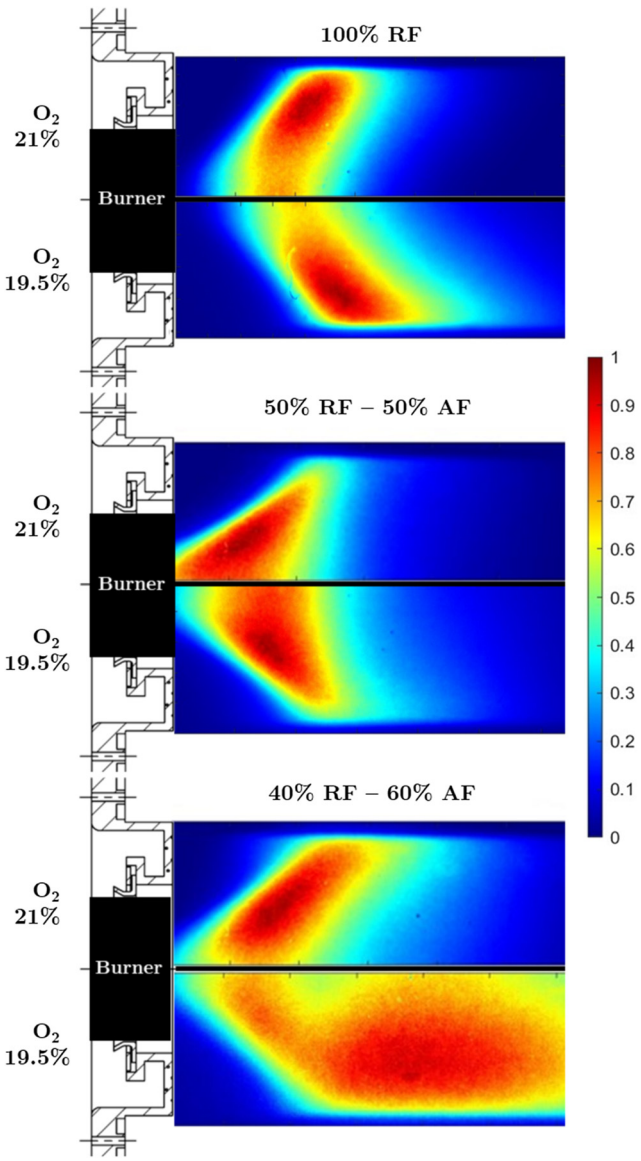

Fig. 12 Comparison between OH* chemiluminescence measurements at various fuel split in normal and vitiated air conditions

enormous difference in terms of chemiluminescence intensity. Indeed, as the available oxygen diminishes, the chemiluminescence intensity significantly decreases for all fuel splits. An almost linear dependence can be identified between the average OH* intensity and the oxygen content. As expected, the limited amount of oxygen available for the combustion process reduces dramatically the mixture reactivity.

For a better comparison, in Fig. 12, the normalized chemiluminescence measurements in standard air condition and vitiated air for each tested fuel flow split are reported. Only the 19.5% oxygen molar concentration maps are presented for the sake of brevity, being a representative example of the oxygen depletion effects on the flame behavior.

This time the maps were normalized each one with its maximum to define more clearly the combustion reaction zone.

Analyzing the 100% RF fuel flow split, it is clearly visible that the flame stabilizes further downstream in the CO_2 addition case. This implies a lower reactivity that causes a slowdown in the chemical reaction process. For the same reason, in terms of shape, the flame seems a little longer with no additional major differences. The same conclusions can be drawn looking at the central image where the two conditions for the 50% RF–50% AF case are reported. Small differences can be appreciated in terms of flame shape. With regards to the last analyzed split, which was selected as the most promising configuration in terms of emissions during the burner characterization, the flame morphology experiences a radical change. The combined effect of the axial fuel line increase and the oxygen content depletion leads to a slowdown of the reaction process. Indeed, the reaction zone becomes more widespread, reaching an extension that almost covers the whole quartz combustion chamber.

In addition to the time-averaged maps of OH* chemiluminescence distribution, some considerations can also be made about the flame unsteadiness. Indeed, the standard deviation of the OH* intensity over time was normalized for each pixel with the local time-averaged value, in order to compare different flow conditions. The thus obtained spatial distribution is showed In Fig. 13 for some representative cases. Figure 13(*a*) corresponds to case 2, where only the radial fuel line is fed. The area where the flame is stable (low normalized standard deviation values) is short and compact, while significant oscillations are present both in the downstream region and at the flame root. Moving to case 5 (Fig. 13(*b*)), where the axial line is also fed, fluctuations are significantly reduced, especially in the downstream region. The effect of CO_2 addition can be analyzed looking at Fig. 13(*c*), which represent the normalized standard deviation of case 11. In this case, oscillations are amplified at the base of the flame, while the normalized standard deviation are just slightly higher in the downstream region than the corresponding case in standard air conditions.

Emissions. Pollutant emissions were measured as well to assess the impact of CO_2 addition and the potential emission abatement linked to the application of EGR to gas turbine engines. For the sake of clarity, only the two extreme fuel line split cases are reported, namely, the 100% RF split and the 40% RF–60% AF fuel split case, to fully characterize the burner behavior. In this case data have been reported as the raw measured value and not corrected at 15% of O_2 at the exhaust, because the O_2 content of the inlet comburent air is different for each case, and such comparison could lead to erroneous conclusions.

Figure 14 shows the CO emissions as a function of the oxygen molar concentration in the incoming mixture for the 100% RF fuel split (cases 7–9) and the 40% RF–60% AF fuel line split (cases 14–17). In a similar way, Fig. 15 reports the corrected NO_x emissions for the corresponding cases. In all cases, the measurements obtained in normal air conditions (21% oxygen molar fraction) are reported as well for a clearer comparison with the vitiated air cases.

Starting from CO emissions, as the available oxygen diminishes the amount of the CO in the exhaust increases. This is one of the main drawback of EGR and one of the main limitations in terms of EGR applicability, apart from the flame stability. Indeed, the low

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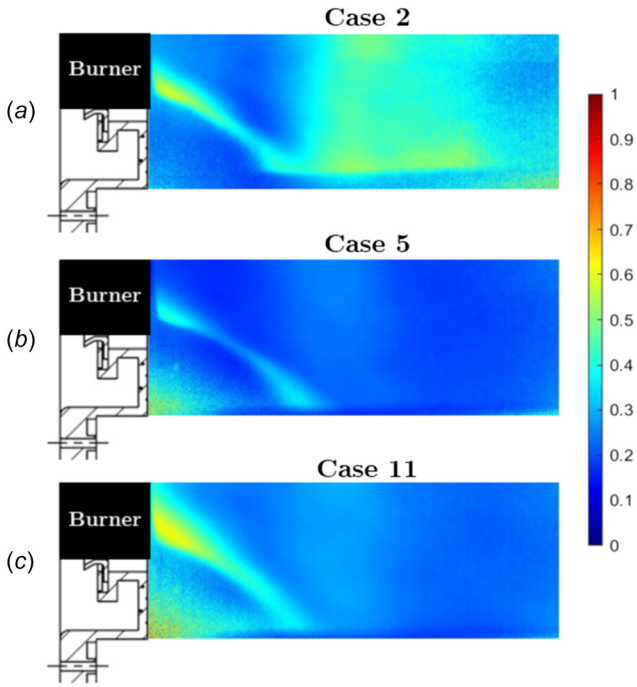

Fig. 13 Maps of the normalized standard deviation of case 2 (*a*), case 5 (*b*), and case 11 (*c*)

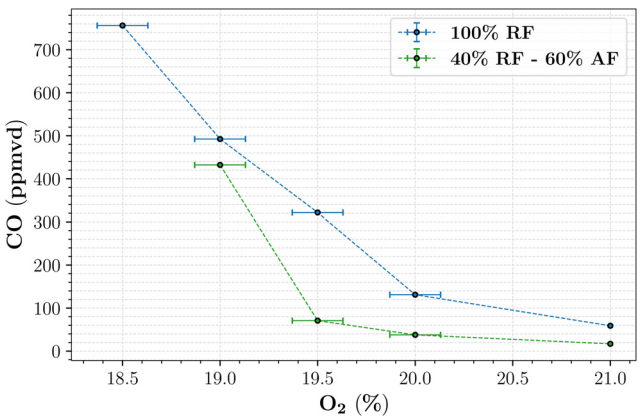

Fig. 14 CO emissions as a function of the oxygen molar concentration for 100% RF and 40% RF - 60% AF fuel split case at constant flame temperature ($T_{flame}^* = 1$)

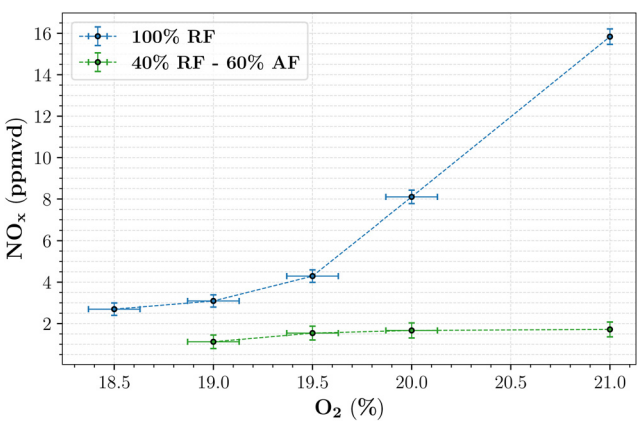

Fig. 15 NO_x emissions as a function of the oxygen molar concentration for 100% RF and 40% RF–60% AF fuel split case at constant flame temperature ($T_{flame}^* = 1$)

availability of oxygen slows down the reactivity and the thermodynamic equilibrium shifts toward a higher CO production. The introduction of the axial fuel line helps to tackle this issue as long as the injected CO_2 amount is moderate. The CO rapidly increases in the 60% AF case at 19.5% oxygen molar concentration due to the lower oxidation rate, while in the 100% RF case the CO emissions continuously grow.

An opposite trend can be found analyzing NO_x emissions (Fig. 15). As pointed out also during the analysis of the OH radical distribution, the CO_2 injection leads to a lower reactivity that in turns helps reducing NO_x emissions. This favorable effect is particularly evident at 100% RF where the NO_x emissions were found to be particularly high compared to the other tested splits because of the poorer mixing. For the 40% RF–60% AF split case, the benefits in terms of NO_x emissions at decreasing oxygen availability are less evident. As pointed out before, the axial fuel line enhances the mixing and drives the NO_x abatement, overshadowing the effect related to the decrease in oxygen content.

Conclusions

A novel lean premixed burner for industrial applications developed by Baker Hughes has been thoroughly characterized with combustion diagnostic techniques, emissions and stability limits measurements in order to explore the feasibility of integrating Exhaust Gas Recirculation.

The burner flow field has been studied through PIV measurements in both reactive and nonreactive conditions, and the typical recirculation zones of swirl stabilized flames are clearly identified. Thanks to OH* chemilumenescence measurements, when the solely radial line is fed, a shift in the flame position is observed as the flame temperature increases: the higher jet penetration moves the flame stabilization region further downstream the burner exit section and the flame assumes a lifted shape. Similarly, at constant flame temperature, changes in the fuel split affect the flame anchoring position: at high and low radial content the flame is lifted, while at more balanced splits the flame anchors in the recirculation zone inside the burner dump region. However, in terms of emissions, both CO and NO_x benefit from an increase in the axial line content. On the contrary, the burner stability limits worsen as the radial line content is decreased, being responsible for the flame stabilization process. As expected, the progressive introduction of the axial line allows a more uniform fuel distribution and a reduction of temperature peaks.

The burner was then tested in vitiated air conditions, at increasing CO_2 content. Chemiluminescence intensity drops as the oxygen content decreases, indicating a lower flame reactivity. As expected, the burner performance in terms of stability worsens as the oxygen available for the combustion process decreases. In vitiated air conditions, the reaction process is globally slowed down and the flame position shifts downstream while maintaining the same shape. However, at 40% RF–60% AF fuel split, the combined effect of the axial fuel line increase and the oxygen decrease leads to an abrupt change in the flame shape: the reaction zone is widespread involving almost the entire liner. In terms of emissions, one of the major drawback of air vitiation is the increase in CO emissions. As the oxidizer oxygen content decreases, the CO amount increases accordingly and a benefit from the 40% RF–60% AF fuel split is still found. However, around 19% oxygen molar fraction, the amount of CO emissions of the aforementioned fuel split almost matches the 100% RF case. A similar behavior is found when analyzing the NO_x emissions trends. On the contrary, the 100% RF case strongly benefits from an oxidizer oxygen decrease. Therefore, a fuel flow split more biased toward the radial line could help the flame stability while not leading to such an increase in terms of emissions.

At the most favorable conditions in terms of NO_x emissions in standard air conditions, the potential emission abatement of close to EGR conditions was addressed in order to define a baseline for further burner development. Although for the baseline only CO_2 was used instead of a realistic EGR mixture, future development will

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benefit from a deeper investigation on the implications of its employment compared to N_2 rich mixtures. In particular, the focus will be on the impact that the different mixtures have on the flame reactivity.

Finally, the present study remarks the importance of dedicated experimental investigations to understand the flame behavior and to have a better overview of new burner concept performance in terms of emissions and stability limits. Chemiluminescence, being relatively simple to implement, bares the potential for a flame diagnostics able to drive the burner design from the early phases of the design process.

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Nomenclature

Acronyms

AA = axial air
AF = axial fuel
CCD = charge-coupled device
CCS = carbon capture and storage
CCU = carbon capture and utilization
CFD = computational fluid dynamics
DLN = dry low NO_x
EGR = exhaust gas recirculation
FoV = field of view
GT = gas turbine
IRZ = inner recirculation zone
ISL = inner shear layer
LBO = lean blow out
LES = large eddy simulations
ORZ = outer recirculation zone
OSL = outer shear layer
PIV = particle image velocimetry
RA = radial air
RANS = Reynolds-averaged Navier Stokes
RF = radial fuel
UHC = unburnt hydrocarbons
VB = vortex breakdown

Symbols

$\Delta P/P$ = combustor pressure drop (%)
P_i = combustor air inlet pressure (Pa)
T_i = combustor air inlet temperature (°C)
T_{flame} = flame temperature (°C)
$T_{\text{flame,ISO}}$ = flame temperature at ISO base load (°C)
$T_{\text{flame}}*$ = normalized flame temperature

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